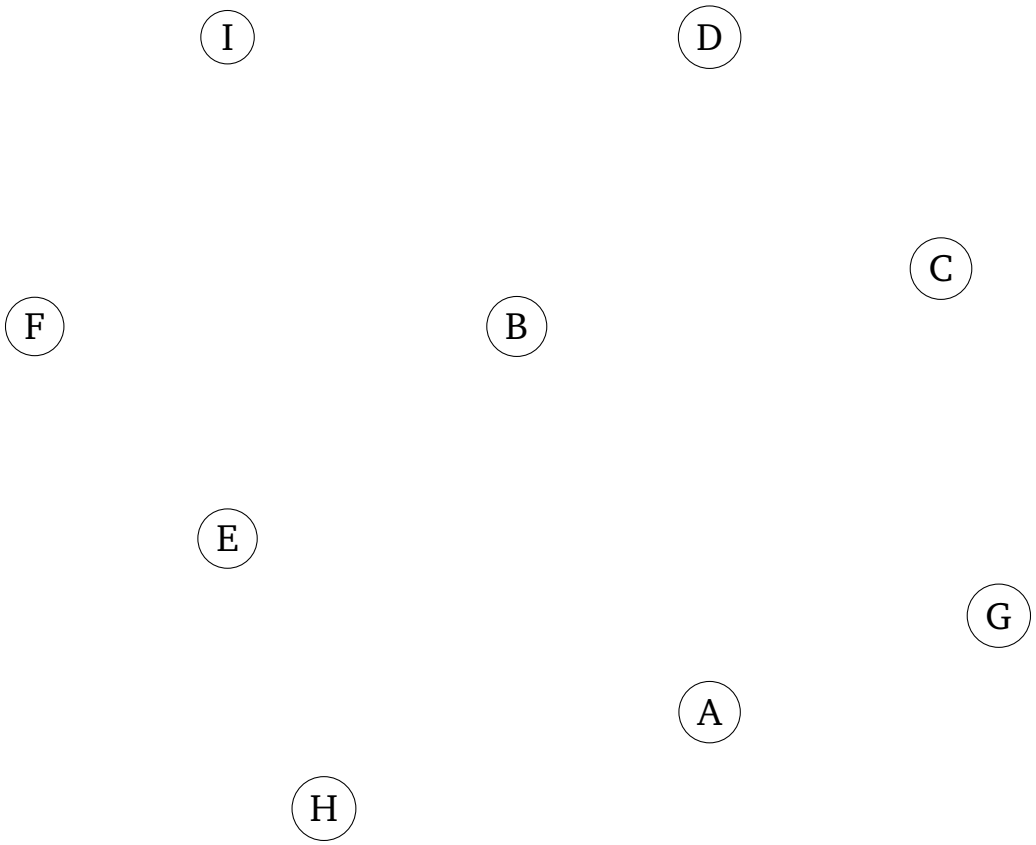
## Part 1 — Nine stations and no money

**Question 1**: Let us design a small power grid network with nine stations A–I (nodes). Initially, these stations are isolated. The cost associated with each connection is provided in the table. Minimize the total cost of the lines to build a single connected network.
Draw your lines on the nine stations below.

| | A | B | C | D | E | F | G | H | I |
|---|---|---|---|---|---|---|---|---|---|
| A | 0 | 2 | 8 | 12 | 4 | 10 | 14 | 6 | 16 |
| B | - | 0 | 3 | 2 | 1 | 5 | 11 | 15 | 7 |
| C | - | - | 0 | 4 | 10 | 14 | 6 | 12 | 16 |
| D | - | - | - | 0 | 5 | 11 | 15 | 7 | 13 |
| E | - | - | - | - | 0 | 6 | 12 | 16 | 8 |
| F | - | - | - | - | - | 0 | 7 | 13 | 9 |
| G | - | - | - | - | - | - | 0 | 8 | 10 |
| H | - | - | - | - | - | - | - | 0 | 11 |
| I | - | - | - | - | - | - | - | - | 0 |




**Question 2**: Compare your grid with your neighbour's. In what order did each of you add the lines? Did the two of you end up paying the same total cost?

**Question 3**: Pick any two stations in your grid. How many different routes run between them?

## Part 2 — Break it

**Question 4**: Consider the scenario where a single station fails and disconnects from the network. Enumerate all possible cases, and compute the size of the largest connected component in each case. Write the sizes in the middle column.

| Station that fails | Size of the largest connected component | That size divided by 9 |
|---|---|---|
| A | | |
| B | | |
| C | | |
| D | | |
| E | | |
| F | | |
| G | | |
| H | | |
| I | | |

**Question 5**: Divide each of those sizes by 9 and write the result in the last column. That fraction is called the **connectivity** of the damaged grid. On average, how much connectivity remains?

**Question 6**: A power-grid network is subjected to a *targeted attack* that breaks the network by removing one node. The attacker targets a single station whose removal minimizes the size of the largest connected component in the resulting network. Which station is most susceptible to this attack?

## Part 3 — One at a time

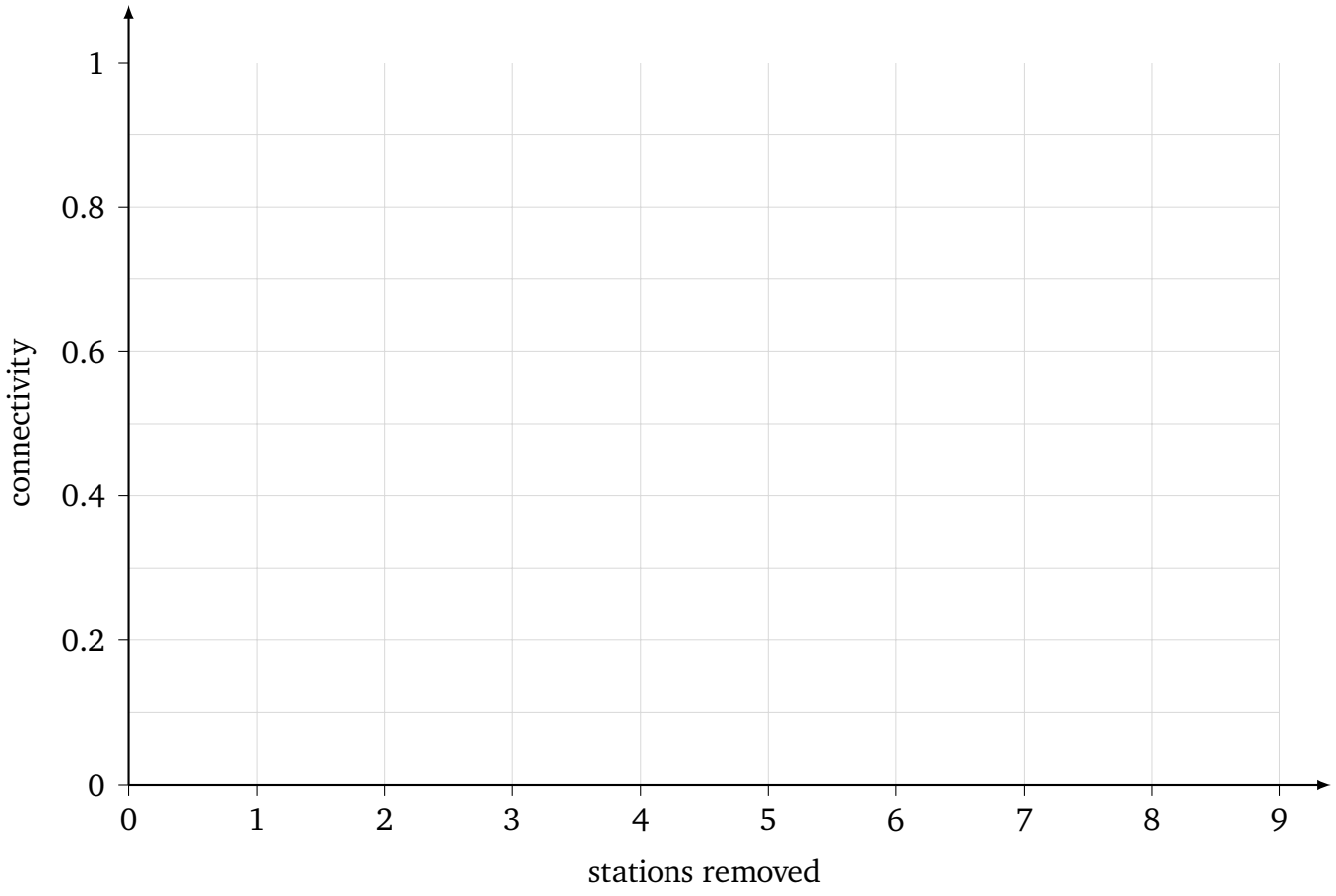
**Question 7**: Now the stations go one after another, and two orders are printed below. Take your grid from Question 1, remove the stations in the given order, and after every removal write the connectivity into the table.

**Order R** (an accident): I, C, H, E, A, G, D, F, B

**Order T** (the attacker): B, A, C, D, E, F, G, H, I

| stations removed | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|
| Order R | 1 | | | | | | | | |
| Order T | 1 | | | | | | | | |

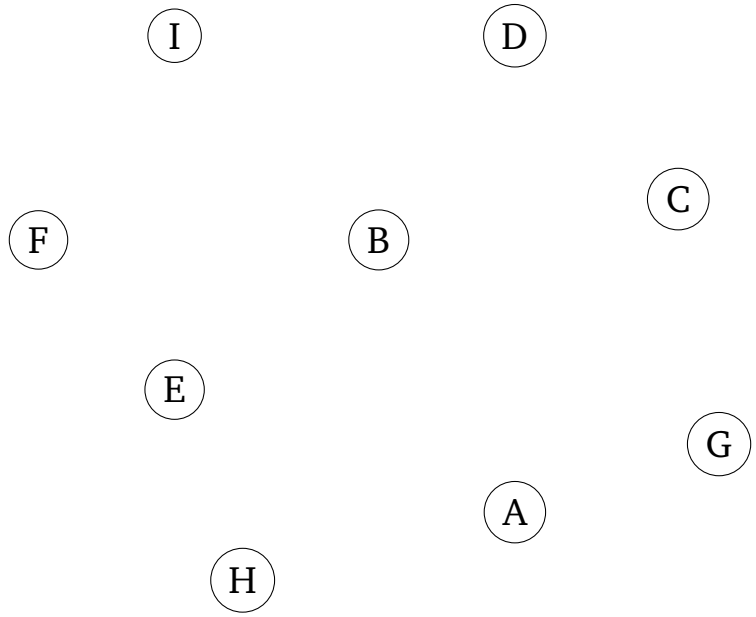
Then plot both rows on the grid, and join each one into a curve.

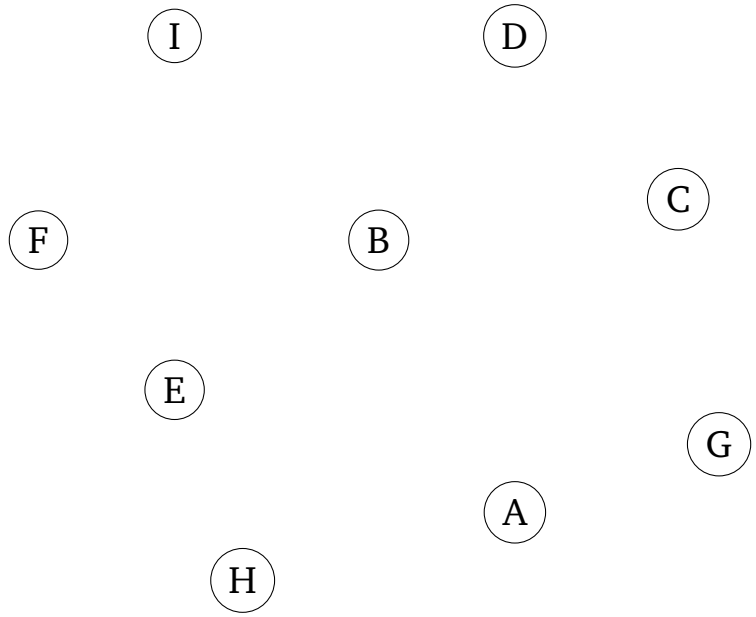



**Question 8**: The area under a curve is a single number for how much of the grid survived the whole sequence. Estimate that area for each of your two curves.

## Part 4 — Build it back

**Question 9**: Redesign the network so that it is as strong as possible against targeted attacks. To simplify, ignore the cost of the lines but keep the same number of lines as your grid in Question 1. You can change the pre-existing lines completely. The attacker can remove only one station.




**Question 10**: Now the attacker can remove two stations at once, and you have additional budget to add four extra lines to protect the network. Design a network of **12** edges so that it is as strong as possible against the targeted attacks.